

Equivalence of DFA & NFA



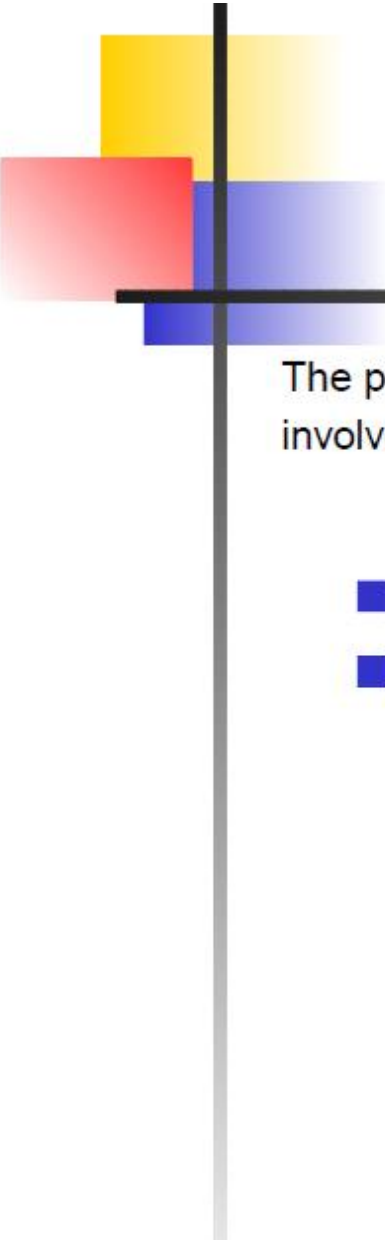
Nondeterministic Finite Automata (NFA)

- A NFA has the power to be in several states at once
- This ability is often expressed as an ability to “guess” something about its input
- Each NFA accepts a language that is also accepted by some DFA
- NFA are often more succinct and easier than DFAs
- We can always convert an NFA to a DFA, but the latter may have exponentially more states than the NFA (a rare case)
- The difference between the DFA and the NFA is the type of transition function δ
 - For a NFA δ is a function that takes a state and input symbol as arguments (like the DFA transition function), but returns a set of zero or more states (rather than returning exactly one state, as the DFA must)



Equivalence of Deterministic and Nondeterministic Finite Automata

- Every language that can be described by some NFA can also be described by some DFA.
- The DFA in practice has about as many states as the NFA, although it has more transitions.
- In the worst case, the smallest DFA can have 2^n (for a smallest NFA with n state).

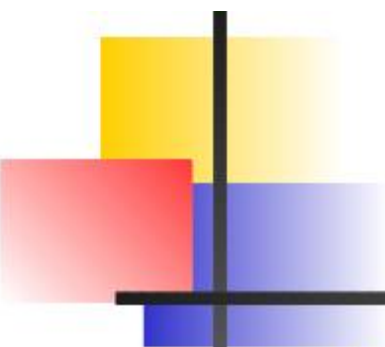


Proof: DFA can do whatever NFA can do

The proof involves an important construction called **subset construction** because it involves constructing all subsets of the set of states of NFA.

From NFA to DFA

- We have a NFA $N = (Q_N, \Sigma, \delta_N, q_0, F_N)$
- The goal is the construction of a DFA $D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$ such that $L(D) = L(N)$.



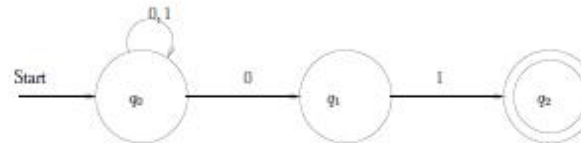
Subset Construction

- Input alphabets are the same.
- The start set in D is the set containing only the start state of N .
- Q_D is the set of subsets of Q_N , i.e., Q_D is the power set of Q_N .
If Q_N has n states Q_D will have 2^n states. Often, not all of these states are accessible from the start state.
- F_D is the set of subsets S of Q_N such that $S \cap F_N \neq \emptyset$. That is, F_D is all sets of N 's states that include at least one accepting state of N .
- For each set $S \subseteq Q_N$ and for each input symbol $a \in \Sigma$

$$\delta_D(S, a) = \bigcup_{p \in S} \delta_N(p, a)$$

To compute $\delta_D(S, a)$, we look at all the states p in S , see what states N goes from p on input a , and take the union of all those states.

Example



$Q_N = \{q_0, q_1, q_2\}$ then $Q_D = \{\emptyset, \{q_0\}, \{q_1\}, \{q_2\}, \{q_0, q_1\} \dots\}$, i.e., Q_D has 8 states (each one corresponding to a subset of Q_N)

	0	1
$\emptyset$	$\emptyset$	$\emptyset$
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_1\}$	$\emptyset$	$\{q_2\}$
$\star \{q_2\}$	$\emptyset$	$\emptyset$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$\star \{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\star \{q_1, q_2\}$	$\emptyset$	$\{q_2\}$
$\star \{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$

Example

